

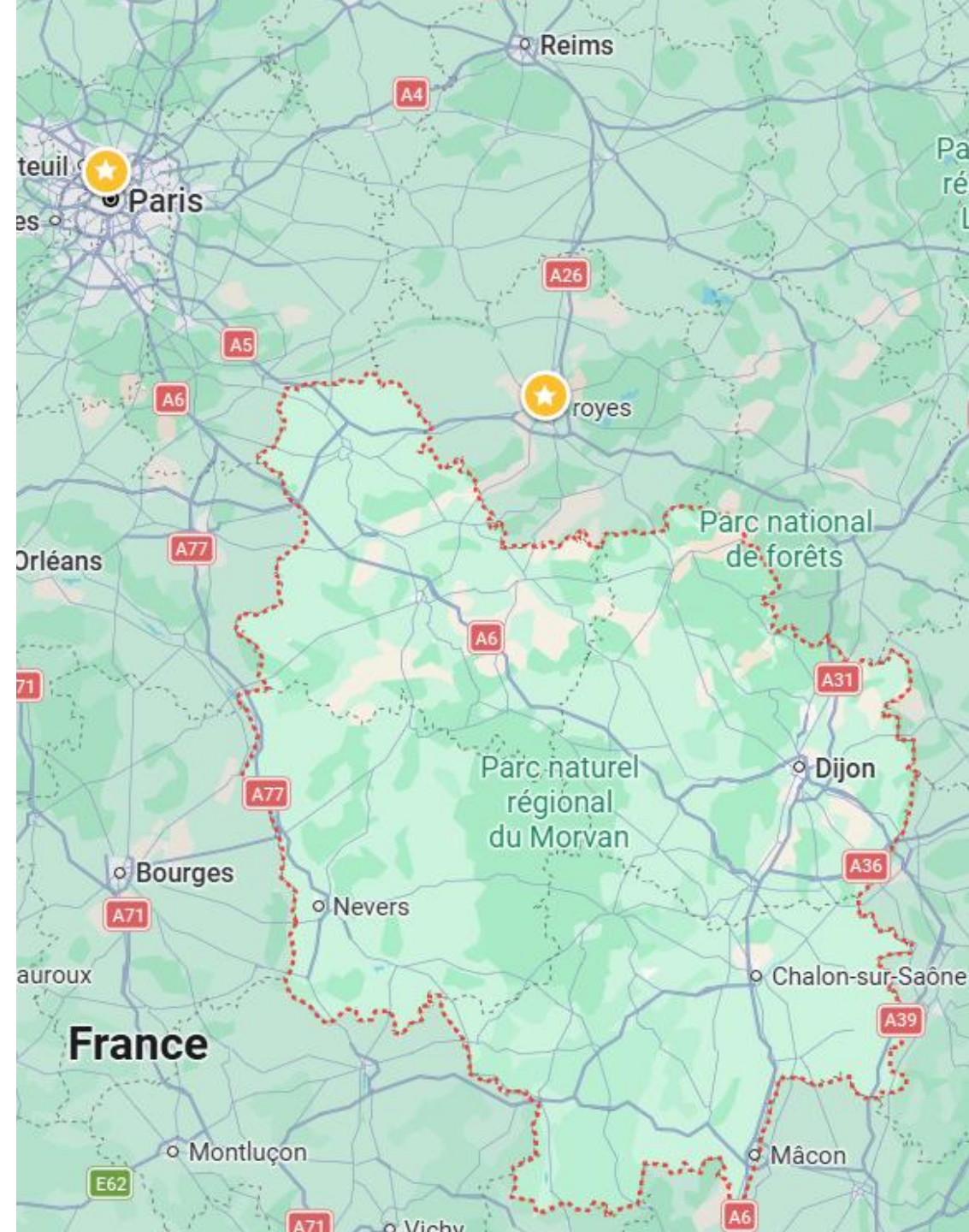
How do I motivate students?

Jérôme Landré - 2026-09-16



Outline

1. Who am I?
2. Teaching and Learning
3. Simple principles
4. My way
 1. Plan activities
 2. Present the goals and ILOs
 3. Present the organisation
 4. Present the team
 5. Present the grades/examination
 6. Define rules
 7. Learning activities
 1. Lectures
 2. Workshops
 3. Labs
 8. The use of AI
 9. Canvas
5. Conclusion



1. WHO AM I?

Jérôme Landré
Assistant Professor
Ph.D. in Image Processing
Born in 1972
Car (number/license) plate collector



Career

Jönköping University (4 years)
Université de Reims Champagne-Ardenne (15 years)
Université de Bourgogne (10 years)

Research

Image Processing and Computer Vision
Applications to medical images, satellite images, patrimonial images...

Teaching

Web development (HTML, CSS, JS, NodeJS, React JS)
Operating Systems (Linux & Windows)
Databases (SQL, NoSQL)

STOP

Didn't you remark anything weird in the previous slide?

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AND THE QUESTION IS



?

BECAUSE, AT FIRST, IT'S



OCH DET ÄR



BECAUSE IT'S

**A WAY TO KEEP STUDENTS'
ATTENTION
DURING THE LECTURES...**

0. WHO AM I?



Assistant Professor



in Image Processing

Born in 1972

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0. WHO AM I?



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in Image Processing

Born in 1972

Car (number/license)



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University of
Ardenn

University
(years)

and Computer

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satellite images, patrimonial
images...

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NodeJS, React JS)

Operating Systems (Linux &
Windows)

Databases (SQL, NoSQL)

**Made
in
France**

FÖRSTA HJÄLPEN
OCH HJÄRT-LUNGRÄDDNING

KOMPETENSOMRÅDET

HLR  rådet



Jag trivs i Sverige

E 3418

Jérôme Landré

Universitetslektor/Assistant Professor
jerome.landre@ju.se

1. Teaching and Learning

Teaching

Teacher's responsibility

- To deliver knowledge
- To create learning activities
- To evaluate the students' examination

Learning

Student's responsibility

- To acquire knowledge
- To participate to the learning activities
- To succeed the course

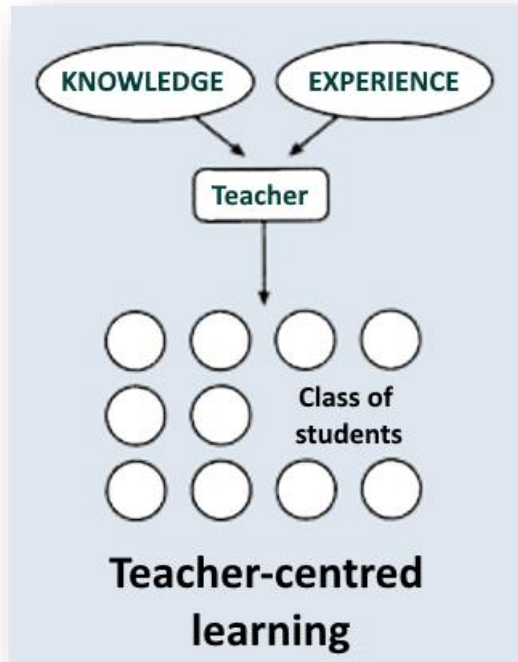
So, a course is 50% Teaching and 50% Learning

It is a **teamwork**: you teach, students learn.

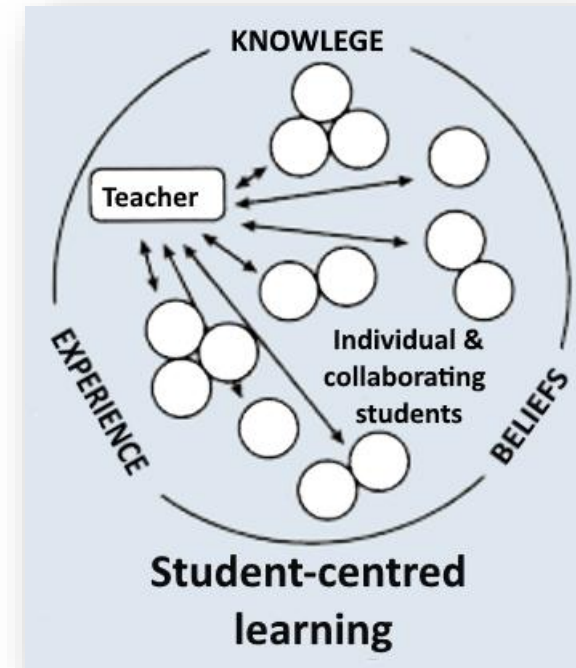
=> Make the students **responsible** for **their** learning!

1. Teaching and Learning

All the students have **the same Knowledge/background**



Real life: All the students are **different/UNIQUE**



2. Simple principles

1. Read articles

SEVEN PRINCIPLES FOR GOOD PRACTICE IN UNDERGRADUATE EDUCATION

*by Arthur W. Chickering
and Zelda F. Gamson**

(from 1987)

Good practice in undergraduate education:

1. Encourages contacts between students and faculty.
2. Develops reciprocity and cooperation among students.
3. Uses active learning techniques.
4. Gives prompt feedback.
5. Emphasizes time on task.
6. Communicates high expectations.
7. Respects diverse talents and ways of learning.

2. Adapt them to your needs: be flexible!

2. Simple principes

*La motivation : condition au plaisir d'apprendre et d'enseigner
en contexte scolaire*

3^e congrès des chercheurs en Éducation
Bruxelles, mars 2004

Rolland Viau
Université de Sherbrooke (Québec)

TABLEAU I : POURQUOI CERTAINS ÉLÈVES SONT MOTIVÉS ALORS QUE D'AUTRES NE LE SONT PAS?

UN ÉLÈVE EST MOTIVÉ SI :	UN ÉLÈVE EST DÉMOTIVÉ SI :
— il considère la matière et les activités qui lui sont proposées <i>utiles</i> ou <i>intéressantes</i>; - et — se sent <i>capable</i> de faire ce qu'on lui demande; - et — a l'impression qu'il a une <i>certaine part</i> de responsabilité (contrôle) dans le déroulement de ses apprentissages et croit qu'il est en grande partie responsable de ses succès comme de ses échecs.	— il considère la matière et les activités qui lui sont proposées <i>inutiles</i> ou <i>inintéressantes</i>; - et/ou — se sent <i>incapable</i> de faire ce qu'on lui demande ou doute qu'il en sera capable; - et/ou — a l'impression de n'avoir <i>aucune</i> responsabilité dans ce qu'on lui demande de faire et croit que ses succès ou ses échecs ne dépendent pas de lui.

2. Simple principles

TABLE I: WHY ARE SOME STUDENTS MOTIVATED WHILE OTHER ARE NOT?

*La motivation : condition au plaisir d'apprendre et d'enseigner
en contexte scolaire*

3^e congrès des chercheurs en Éducation
Bruxelles, mars 2004

Rolland Viau
Université de Sherbrooke (Québec)

A STUDENT IS **MOTIVATED** IF HE/SHE:

finds the proposed material and activities
useful or interesting;

and

feels capable of doing what is asked;

and

feels he/she has some responsibility (control)
in the conduct of his/her learning and believes
he/she is largely responsible for his/her
successes as well as his/her failures.

A STUDENT IS **NOT MOTIVATED** IF HE/SHE:

considers the proposed subject matter and activities
useless or uninteresting;

and/or

feels unable to do what is asked of him or doubts his/her
ability;

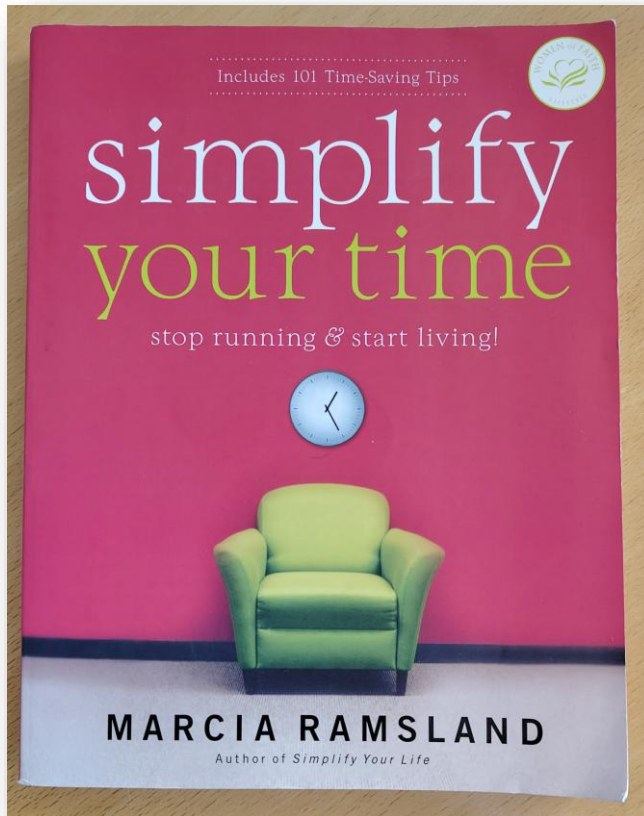
and/or

has the impression of having no responsibility in what
he/she is asked to do and believes that his/her successes
or failures do not depend on him/her.

 => **NO MOTIVATION** comes faster and is easier than **MOTIVATION!!!**

2. Simple principles

Simplify your time!



Time-Saving Tip #2

Remember, time is on your side if you use it well. But it works against you if you bite off more than you can chew.

Time-Saving Tip #3

You can add time to your life by deleting an activity that is unproductive. But you multiply your time when you add an activity that energizes you.

Time-Saving Tip #4

Those who arrive early have more control over how an event will go than the person arriving late. Be punctual!

Time-Saving Tip #44

The more you prioritize, the better you will become at gauging how much you can accomplish. Prioritizing is the best way to sort and simplify a long list.

Time-Saving Tip #7

Two-minute pickups are tasks that are too short to write on a to-do list but that slow the pace of your life when they are ignored too long.

Time-Saving Tip #63

When you're overwhelmed, write a list and prioritize. You will cut the time needed to accomplish your tasks in half if you follow your priorities.

3. My way

Score points with the students

- **One smile = one point**
- Surprise the students
- Use humor (sparsely)
- Make them curious
- Make them have success
- Be holistic: **savoir-faire** et **savoir-être**
- Make a difference
- **Be positive!**

Communication

- Be consistent
- Create your branding
- Answer students' questions
- Answer students' emails
- Send an announcement on Canvas every week to keep them on track
- **Be positive!**

TIPS AND TRICKS

POSITIVE ATTITUDE CAN CHANGE LIFE

Syed Muhammad Sajjad Kabir

Chairman and Assistant Professor, Department of Psychology

BE POSITIVE!

*Remember that **YOU** control your attitude:*

Use a more positive vocabulary:

Be thankful:

Smile:

Understand that we all need and deserve time to recharge:

Follow a healthy lifestyle:

Be open to humor: Give yourself permission to smile or laugh, especially during difficult times. Seek humor in everyday happenings. When you can laugh at life, you feel less stressed.

TIPS AND TRICKS

POSITIVE ATTITUDE CAN CHANGE LIFE

Syed Muhammad Sajjad Kabir

Chairman and Assistant Professor, Department of Psychology

BE POSITIVE!

Be grateful or have gratitude in what you have today. Learn to utilize what you have in your hands and make use of it in the most constructive way.

When you fall, you must understand that everyone falls no matter how great or how stable they may look. When everything seems to fall apart, tell yourself that if you could get out of this mud of sufferings, you could be one of the greatest persons on earth.

Positive attitudes lead to happiness and success and can change your whole life.

TIPS AND TRICKS

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BE POSITIVE!

Be grateful or have gratitude in what you have to
hands and make use of it in the most constructive way

When you fall, you
they may look. When
mud of sufferings

...
We've gotta hold on to what we've got
It doesn't make a difference if we make it or not

...
We'll give it a shot!

...
<https://www.youtube.com/watch?v=IDK9Qqlzhwk>

Positive attitudes lead to happiness and success and can change your whole life.

3. My way

- **Example of a course:**

- TGWK12 - Web Development Fundamentals, 7.5 credits
- **170+ students**
- Two students, Mira POP and I as lab teachers
- 3 programs: Digital Mobile Platforms (**DMP**), Network Infrastructure and CyberSecurity (**NICS**) and New Media Design (**NMD**)
- **Learning activities:**
 - Lectures (Theory/Concepts, Code examples, Live Coding Sessions)
 - Workshops (Active Learning Classrooms, ALC)
 - Labs
 - + Resources to read (web pages and pdf documents) and/or to watch (YouTube videos)



3. My way

1. Plan activities

TGWK12 - Web Development Fundamentals - Autumn 2025										Jönköping University, School of Engineering Computer Science and Informatics Department	
Course organisation											
Week	36		37		38	39	40	41	42	43	
Programme	1 DMP/NICS	1 NMD	2 DMP/NICS	2 NMD	3	4	5	6	7	8	
Lecture 1 1:45	Organisation & HTTP Jérôme		Javascript (client-side) & Git Jérôme	Databases 2 Jérôme	Node.js + Express Jérôme	Express + REST API Jérôme	Forms/authentication/ session/cookies Jérôme	Security/AI tools Jérôme	Advanced Code 1 (pagination) Jérôme	Inspira online, on campus Examination Jérôme	
Lecture 2 1:45	HTML & CSS 1 Jérôme	Databases 1 Jérôme	Project Work Jérôme			Express/Handlebars Jérôme	Git/Databases Jérôme	Databases/CRUD Jérôme	Advanced Code 2 (upload images) Jérôme		
Lecture 3 1:45	HTML & CSS 2 Jérôme										
Workshop (Active Learning Classroom) 1:45		Databases 1 Jérôme	Javascript (client-side) & Git Jérôme	Databases 2 Jérôme	Workshop Project Setup Jérôme						
Lab 3:45	Lab1 HTML/CSS ? + Mira + Jérôme		Lab2 Advanced CSS/JS ? + Mira + Jérôme		Lab3 Node/Express, Git ? + Mira + Jérôme	Lab4 Express/Handlebars & Project Work ? + Mira + Jérôme	Lab5 Databases & Project Work ? + Mira + Jérôme	Lab6 Forms, Sessions, Cookies & Project Work ? + Mira + Jérôme	Lab7 Pagination, Security, Full CRUD & Project Work ? + Mira + Jérôme		
Project			Portfolio or Open Project / Individual or groups of TWO students							Project evaluation Jérôme	
Number of students registered in the course:			160+			People involved in the course:					
Number of lab groups:			6 in total, 2 for Mira, 1 for ?, 1 for ?, 2 for Jérôme			?			1 lab per day, Thursdays and/or Fridays		
Number of needed lab assistant students:			2			?			1 lab per day, Thursdays and/or Fridays		
						Mira Pop			1 lab per day, Thursdays and/or Fridays		
						Jérôme Landré			1 lab per day, Thursdays and/or Fridays		
The course has two 1:45 sessions a week (except first three weeks).											
The course has one 3:45 lab session per week.											
Evaluation:			Inspira, project work evaluation								

Course organisation

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?
?
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1 lab per day, Thursdays and/or Fridays
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The course has two 1:45 sessions a week (except first three weeks).
The course has one 3:45 lab session per week.
Evaluation: Inspira, project work evaluation

3. My way

2. Present the goals & ILOs

1) GOAL OF THE COURSE

Learn the basics of modern web development:

- On the **client** side:
 - Internet & HTTP
 - HTML & CSS
 - Javascript (client-side) aka JS
- On the **server** side
 - Node.js (server-side Javascript)
 - Express
 - Handlebars
 - Databases (SQLite3)
- And several tools/technologies to help your development:
 - JSON
 - CSS Frameworks
 - Git versioning
 - REST API & Postman
 - Code editor: Visual Studio Code

2) ILOS (INTENDED LEARNING OUTCOMES)

- **Knowledge and understanding**
 - display knowledge of the fundamental building blocks of the web (HTTP and HTML)
 - demonstrate comprehension of common security concerns in web applications
- **Skills and abilities**
 - demonstrate the ability to construct web pages using HTML5, CSS3, and CSS frameworks
 - demonstrate skills of programming in JavaScript
 - demonstrate skills of creating web pages using Node.js
 - demonstrate skills of constructing database-backed web applications

3. My way

3. Present the organisation

4) ORGANISATION

- Canvas page of the course:
<https://ju.instructure.com/courses/14847>
- 170+ students:
 - TGM7 Computer Engineering: Software Engineering and Mobile Platforms
 - TGGD7 New Media Design
 - TGNV2 Networking, Infrastructure and Cyber Security
 - Exchange students from Austria, Belgium, China, Italy, Mexico, Spain, Thailand
- 6 lab sessions on Thursdays and Fridays (morning **OR** afternoon). **TO C**
group: find a student in **another group** and send me (**both of you**) and
you want to swap your groups. I will proceed to the change.

4) ORGANISATION

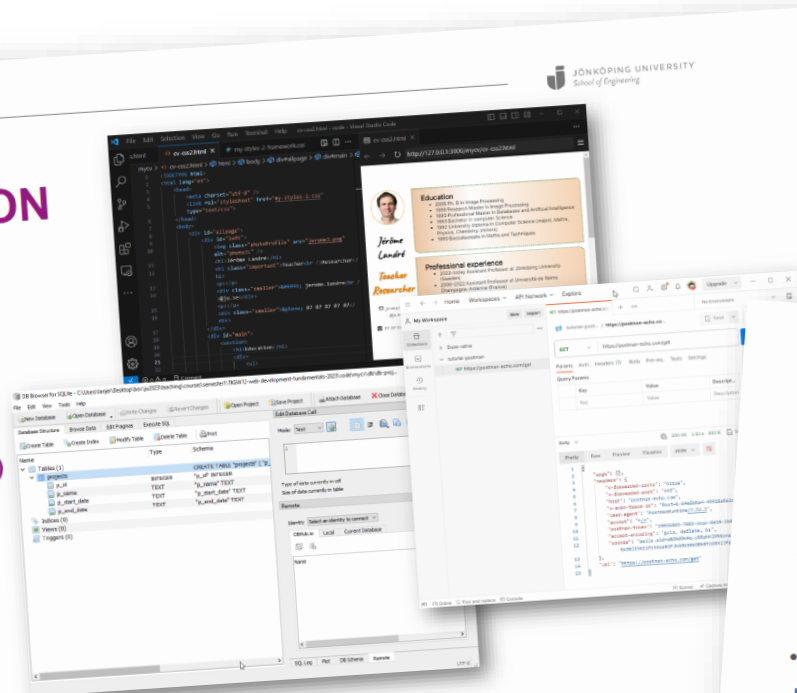
- Learn many concepts in parallel:
 - Internet & HTTP
 - HTML & CSS
 - Javascript
 - Node.js, Express, handlebars
 - Databases (SQL within SQLite3)
 - Git
- **Lecture/workshop/seminar** sessions
- **Lab** work
- **Project** work

3. My way

2. Present the organisation

4) ORGANISATION

- Software needed for this course on a computer:
 - Web Browser
 - Visual Studio Code
 - Node.js & npm
 - (DB Browser for SQLite)
 - Git
 - (Postman)



4) ORGANISATION

- Per week, there are (except weeks 1, 2 and 3):
 - **Two** 1:45 hours lecture/workshop sessions
 - **One** 3:45 hours lab session (on Thursdays or Fridays)
- Online schedule made by Daniel LUNDQVIST, tack!

<https://cloud.timeedit.net/ju/web/staff/riq8vd9Q003Z8YQt67704vZQ6YZZj40jXm0n50kQ71o0wo2nQ5701Z5p00w7uY058Z48Q86E2Q9C78ZQ024tAn4FQ72c829AE08126D215Zq6Qo.html>

- Be careful, **several changes** can occur during the course due to constraints on the teachers, rooms or, other unpredictable events
- You will receive a notification on Canvas if there is any change

3. My way

3. Present the team

4) PRESENTATION OF THE TEAM



4) PRESENTATION OF THE TEAM

Mira POP

Former NMD student, Lab assistant



Jérôme LANDRÉ

Coordinator, Lecturer and Lab assistant



3. My way

4. Present the grades/examination

5) GRADES

- The course is graded 5, 4, 3 or Fail.
- It is divided in two parts:
 - **Theory** (2 credits, graded 5, 4, 3, Fail):
 - Written examination
 - **Practice** (5.5 credits, graded 5, 4, 3, Fail):
 - **Project** work:
 - Web site development
 - More information on the project will come soon...

Registration of examination:

Name of the Test	Value	Grading
Project Work	5.5 credits	5/4/3/U
Written examination	2 credits	5/4/3/U

5) GRADES

- The project work will consist in **building a one-page website** using:
 - Node.js/Express
 - Handlebars
 - HTML5/CSS3
 - Javascript
 - SQLite3 database with at least three tables
 - One or two user(s) authentication
- More information will come soon.

3. My way

4. Present the grades/examination

Written examination

(don't forget to
register for it!)

Web Development Fundamentals 7.5 HP- TGWK12
TGWK12/2202 2 HP (2202 Written examination)
Friday 17 October time 09:00 to 11:00
The registration will open 2025-09-17
Type of exam: Inspira - Digital exam **Anonymous**
Room: **Campus Jönköping**
Free text: Read and find the rules for exams here:
<https://ju.se/student/en/studies/examination.html>

Source: <https://ju.se/student/en/studies/examination/exam-schedule-jth.html>

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Grading criteria for the Web Development Fundamentals project – 2025 – v1.2

Grade	Requirement/Code content for INDIVIDUAL project	Requirement/Code content for GROUPS of 2 students project
3	- The code works without error, there are sufficient comments in the code, - The code uses Node.js, Express, Handlebars and SQLite3, - The final website is a one-page dynamic application, the menu MUST contain at least: Home, List (can be renamed depending on your project choice), About, Contact, - The Database has at least two tables (at least 5 records) that are displayed on the pages, the tables have at least 3 fields, - The menu "List" must link to a page displaying the list of elements (except the user table used for grade 5) from one of your tables, a detailed view of one element is given when you click on it, - Basic security (no SQL injection) and login/encrypted password checked to modify the data, there is a message to know if the user is connected or not, - The login and password for the user are respectively "admin" and "wdf#2025", - Usage of SELECT/INSERT/UPDATE/DELETE (CRUD) operations to display/modify one table, - Acceptable UI/UX using basic CSS.	Requirements are the same as the left column - WITH three major changes in red: - The Database has at least four tables (at least 10 records) that are displayed on the pages, the tables have at least 3 fields, - Usage of SELECT/INSERT/UPDATE/DELETE (CRUD) operations to display/modify two tables,
4	- Requirements for grade 3 above PLUS: - Database with three or more tables (at least 5 records in each table), two of them being linked (SQL INNER JOIN) in at least one query (that is used in the pages), - There is a list with a dynamic pagination system (at least 3 pages with 3 elements per page) for one table (so, at least 9 records in this table) - Moderate UI/UX with Flexbox or Grid CSS.	Requirements are the same as the left column and the cell above WITH three major changes in red: - Database with five or more tables (at least 10 records in each table), three of them being linked (SQL INNER JOIN) in at least one query (that is used in the pages),
5	- Requirement for grade 4 above PLUS: - Create a complete user management system: create new user with hashed password, listing, updating, and deleting users (SELECT/INSERT/UPDATE/DELETE in your users' table), the users can log in on your website, - Advanced UI/UX being fully responsive (computer, tablet, and phone) with Flexbox or Grid CSS or a CSS Framework.	Requirements are the same as the left column and the cell above WITH two major additions in red: - In one of the forms, there is a possibility to upload an image (PNG or JPG) in a directory on the server with SELECT/INSERT/ UPDATE/DELETE on this image which is used and displayed in the web pages,

7) RULES

- Your role as a student:
 - be **engaged**, **active** and **interested** by the course:
 - actively attend the lectures to get the building blocks of knowledge necessary to understand the different concepts
 - actively attend the workshops/seminars/live coding sessions to have examples of applications
 - work and ask for help and feedback during the labs to apply the concepts to real-life professional situations
 - Ask questions to the teachers, ask for help when needed
 - Take into account teachers' feedback to improve the assessments
 - Give feedback on advantages and drawbacks of this course to the coordinator
 - Be **polite** and **respectful**
 - Don't **speak too loud** during lecture/workshop/lab sessions
 - Be **POSITIVE!**

7) RULES

- Our role as teachers:
 - **Facilitate** (from Björn Praestgaard-Larsen) students' learning
 - **Make** the students discover and explore the concepts on their own (with questions and feedback)
 - **Create** a good pedagogical environment for students
 - Be **inclusive** to include every student
 - Be **polite** and **respectful**
 - Be **POSITIVE!**

9) HOW TO SUCCEED IN THIS COURSE?

- **Attend** the lectures!
- **Participate** in the workshops/seminars!
- **Do** the labs, ask questions!
- **Do** the project!
- **Cite** your sources to avoid plagiarism!!!

Code, code and code again...

9) HOW TO SUCCEED IN THIS COURSE?

REMEMBER, it's all about



Source: <https://news.stanford.edu/stories/2018/06/four-ways-human-mind-shapes-reality>

9) HOW TO SUCCEED IN THIS COURSE?

You are the



to your success!

9) HOW TO SUCCEED IN THIS COURSE?

On the road to success, you are your own



9) HOW TO SUCCEED IN THIS COURSE?

Of course the teachers will help you, but we are only facilitators,

YOU DO THE



9) HOW TO SUCCEED IN THIS COURSE?

If you lack knowledge, just fill the



9) HOW TO SUCCEED IN THIS COURSE?

So STAY

READ THE

HAVE

AND BE



- Course given in **English**
- **French** teacher
- In Jönköping, **Sweden**
- Many **International** Students
- Several **programmes** involved

Challenges?

(EDUCATE: Managing the course, Broadened participation)

- Course given in **English**
- **French** teacher
- In Jönköping, **Sweden**
- Many **International** Students
- Several **programmes** involved

Opportunities

(EDUCATE: Managing the course, Broadened participation)

3. My way

6. Learning activities: Lectures

One smile = one point

- Smile when you enter the lecture room: **happiness** is **contagious**
- Use humor
- **Surprise** the students // **ASK** questions (interactions)
- Make a pause in the middle of the lecture: **TA EN**
- Move during the lecture // **ALWAYS** use a microphone (EDUCATE)
- Be consistent on your slides
- Propose "tips and tricks" slides: positivity, music, fun facts, french language...
- **Be positive!**



TIPS AND TRICKS: FRENCH WORDS



Bonjour
Au revoir

Ça va ?
Ça va.

Merci

Oui
Non



Good day
Good bye

How are you?
I'm fine.

Thank you

Yes
No



God dag
Hej då

Hur mår du?
Jag mår bra.

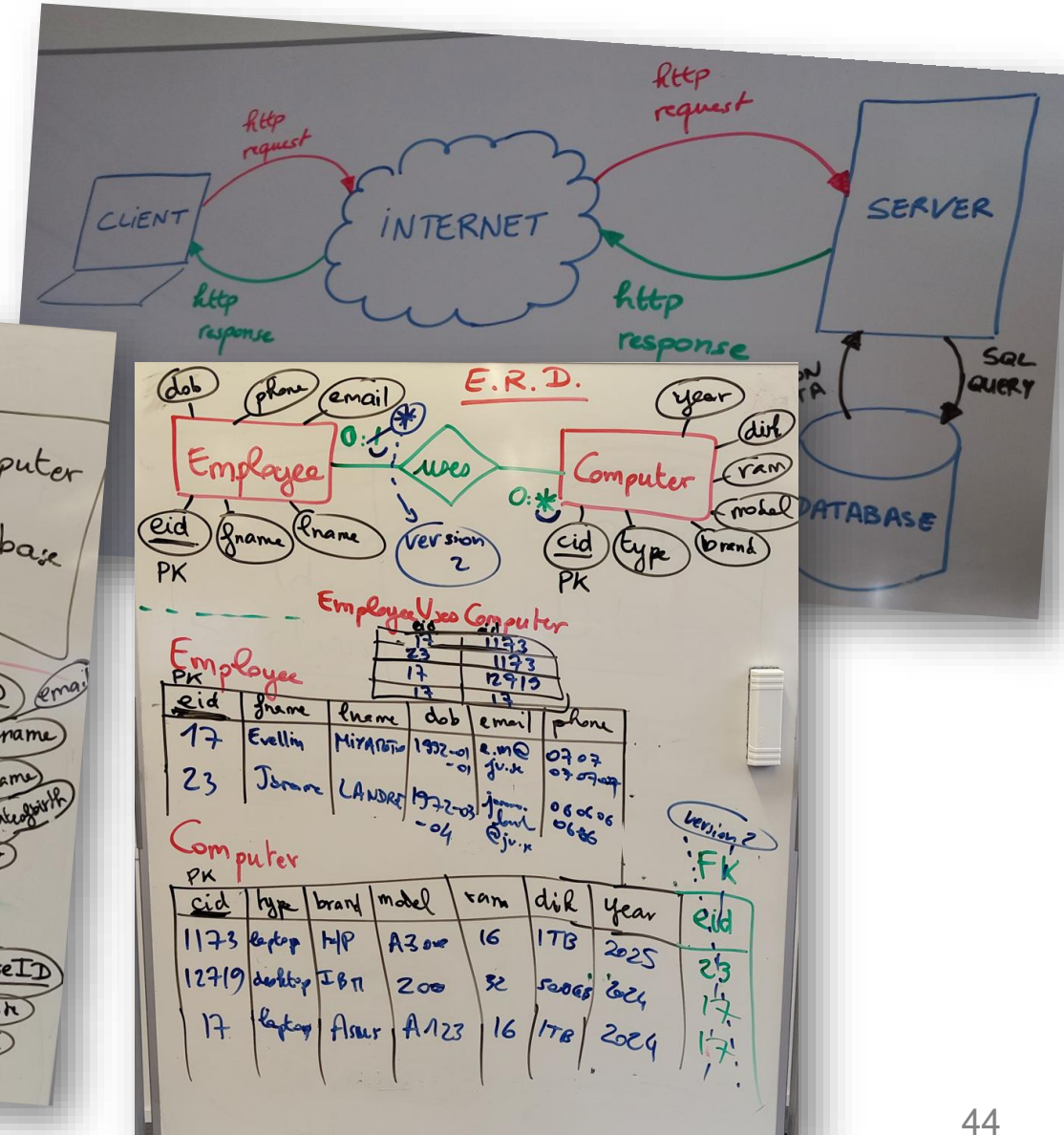
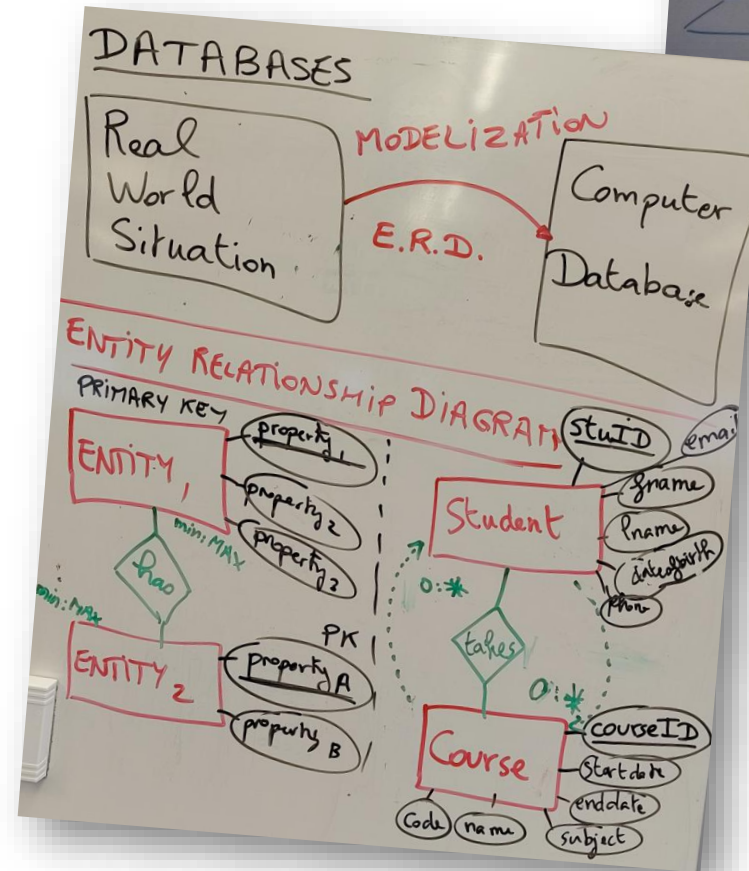
Tack

Ja
Nej

3. My way

6. Learning activities: Lectures

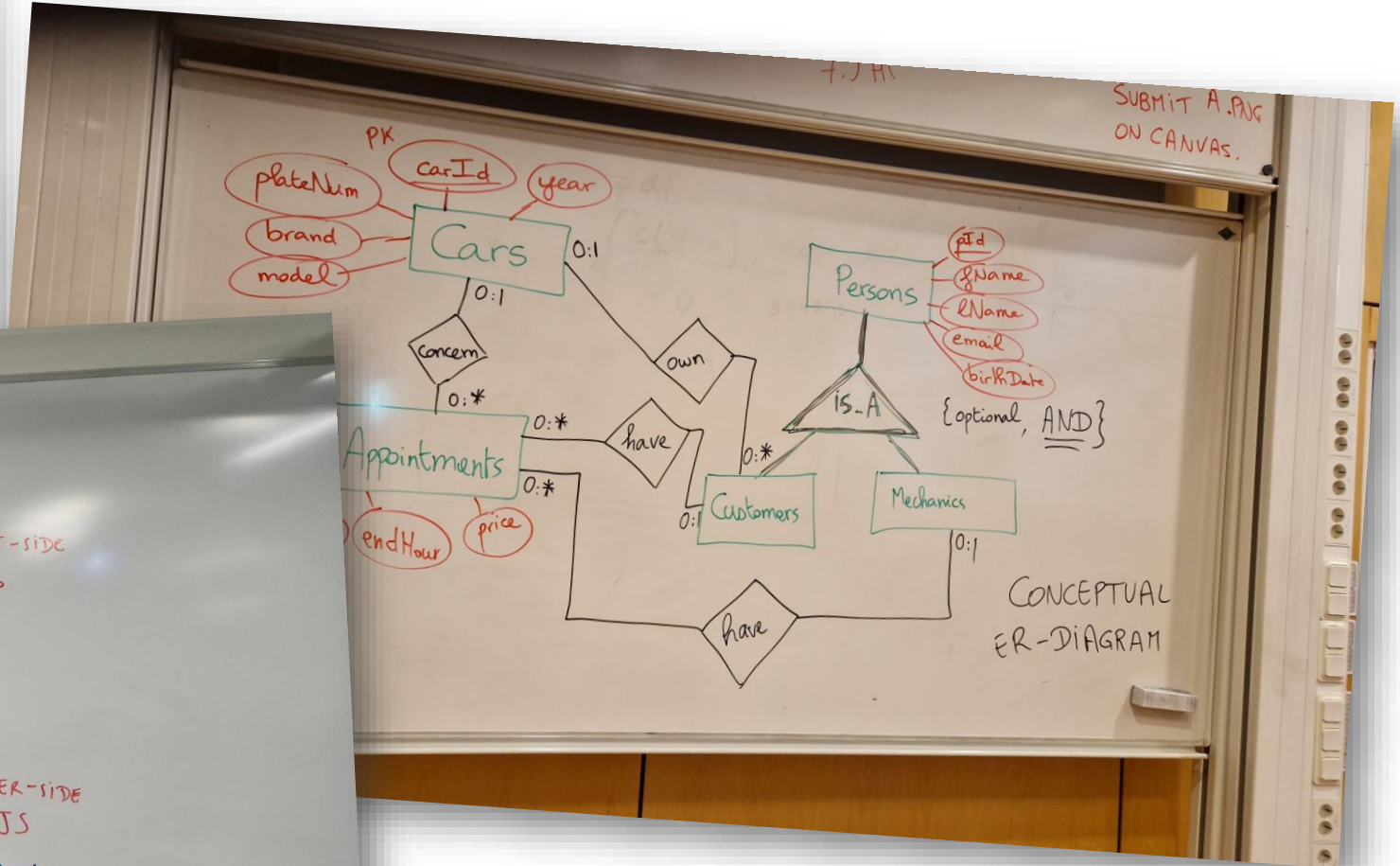
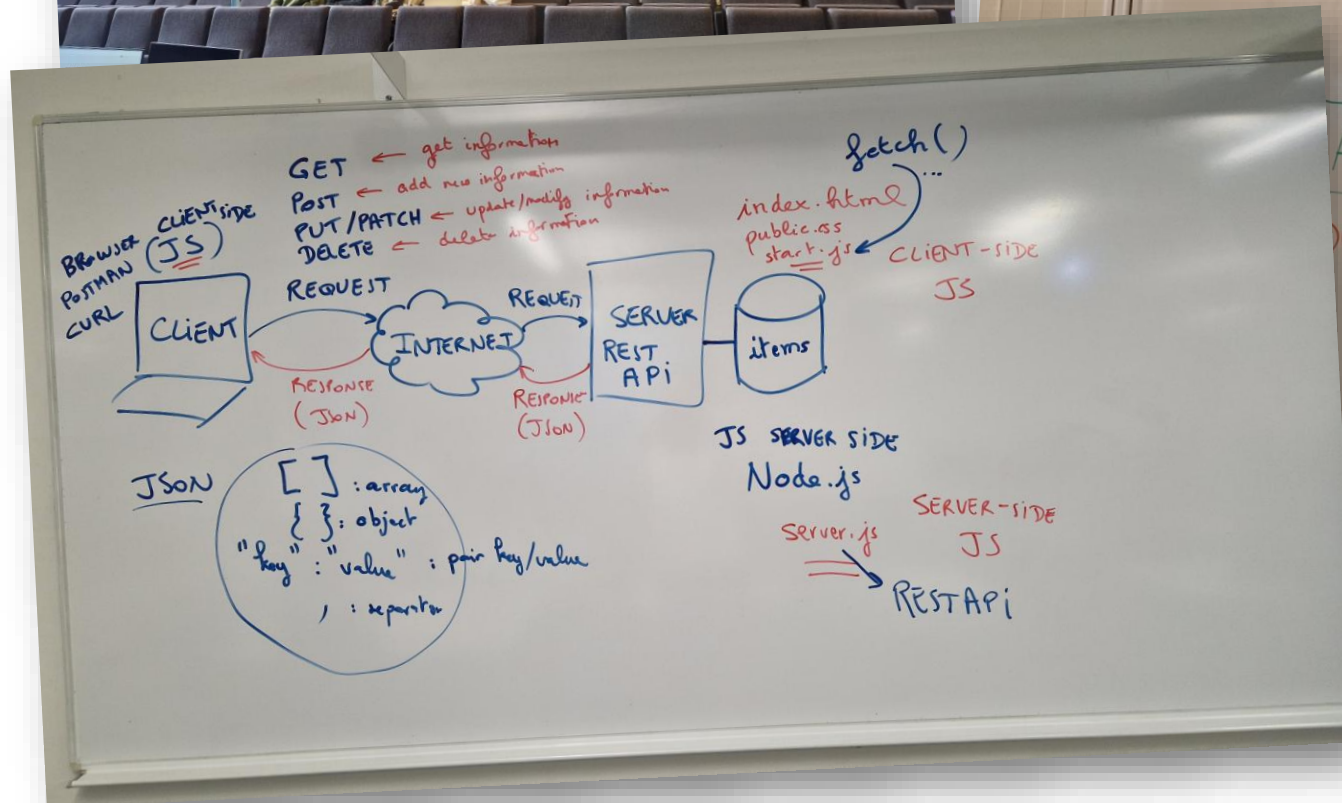
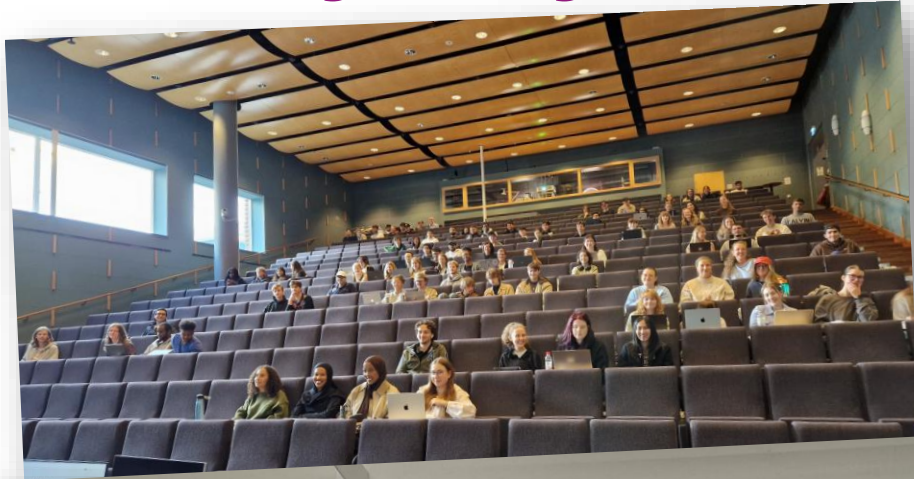
- Powerpoint slides
- Whiteboard (old school)



- Be positive!

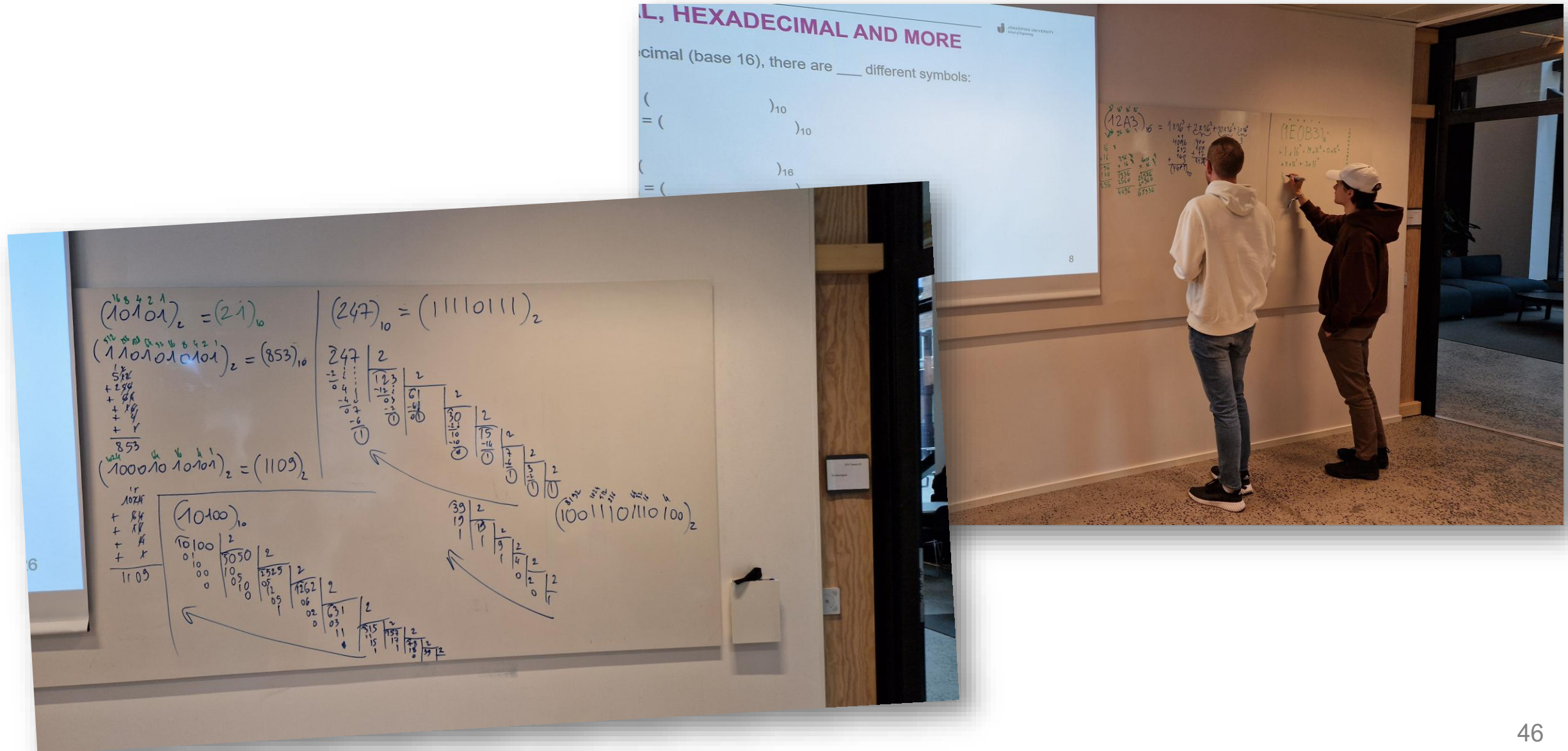
3. My way

6. Learning activities: Lectures



3. My way

6. Learning activities: Lectures



Tips and Tricks

Be creative!

Remember: One smile = one point

You want to catch the students' attention

REMEMBER:

WEB DEVELOPMENT FUNDAMENTALS

WEB DEV IS FUN



3. My way

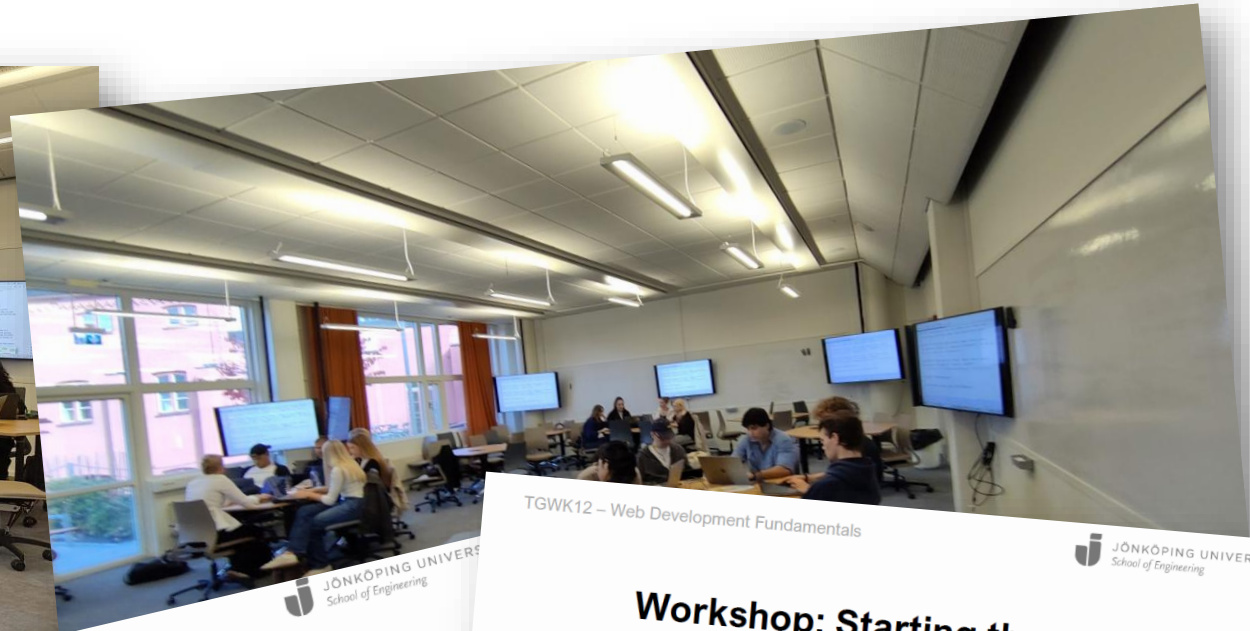
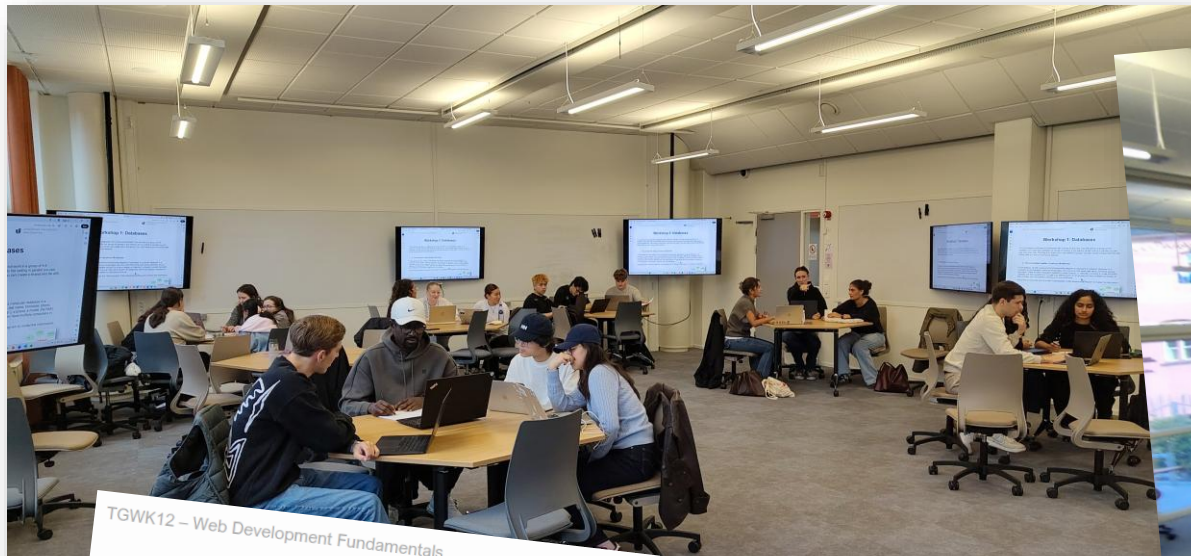
6. Learning activities: Workshops

- Active Learning Classroom (ALC)
- Propose small and progressive exercises
- Make the students discuss around the table (teamwork)
- Make a pause in the middle of the session
- Move during the workshop // **ALWAYS** use a microphone
- Answer questions and give feedback
- **Be positive!**



3. My way

6. Learning activities: Workshops



TGWK12 – Web Development Fundamentals



Workshop 1: Databases

This workshop is dedicated to Databases with a focus on SQLite3. You will work in a group of 4-5 students. You need one computer per group (of course, if you want to do the coding in parallel you can use more than one). The idea is to share your code within the group. You can create a shared text document for all the code you will run during this session.

1) The conceptual model - duration: 20 minutes:

In this session, we will create an Entity-Relationship diagram to represent a computer database in a company. In the company, there are employees, we know their first name, last name, birthdate, phone, and email. There are also computers defined by a type ('laptop' or 'desktop'), a brand, a model, the RAM capacity, the disk capacity and the year they were bought. An employee can have multiple computers in the company but a computer is used by only one employee.

According to this small text, draw the "full/complete" Entity-Relationship Diagram to model the information.

TGWK12 – Web Development Fundamentals



Workshop 2: Databases SQLite3 in Node.js with Express

This workshop is dedicated to Databases with a focus on SQLite3. You will work in a group of 4-5 students. You need one computer per group (of course, if you want to do the coding in parallel you can use more than one). The idea is to share your code within the group. You can create a shared text document for all the code you will run during this session. You MUST use the database you created in the previous workshop.

1) Node and Express - duration: 1,5 hour

TGWK12 – Web Development Fundamentals



Workshop: Starting the project

This workshop is dedicated to starting the project. During this sessions, you will work in a group of 4-5 students. You need one computer per student. The idea is to discuss your code within the group. You can create help the other members of the group to setup their computer to start the project. I recall that the project is made **INDIVIDUALLY** or **WITH A GROUP** of **TWO STUDENTS**.

1) Installing Node.js - duration: 15 minutes:

You must know if you have node installed on your computer: run the command "node -v" in a terminal. If you have node v22.19.0, perfect, node is installed on your system!

If not, please, download and install Node.js on your computer, please use the "Long Term Support (LTS)" version from: <https://nodejs.org/en/download>

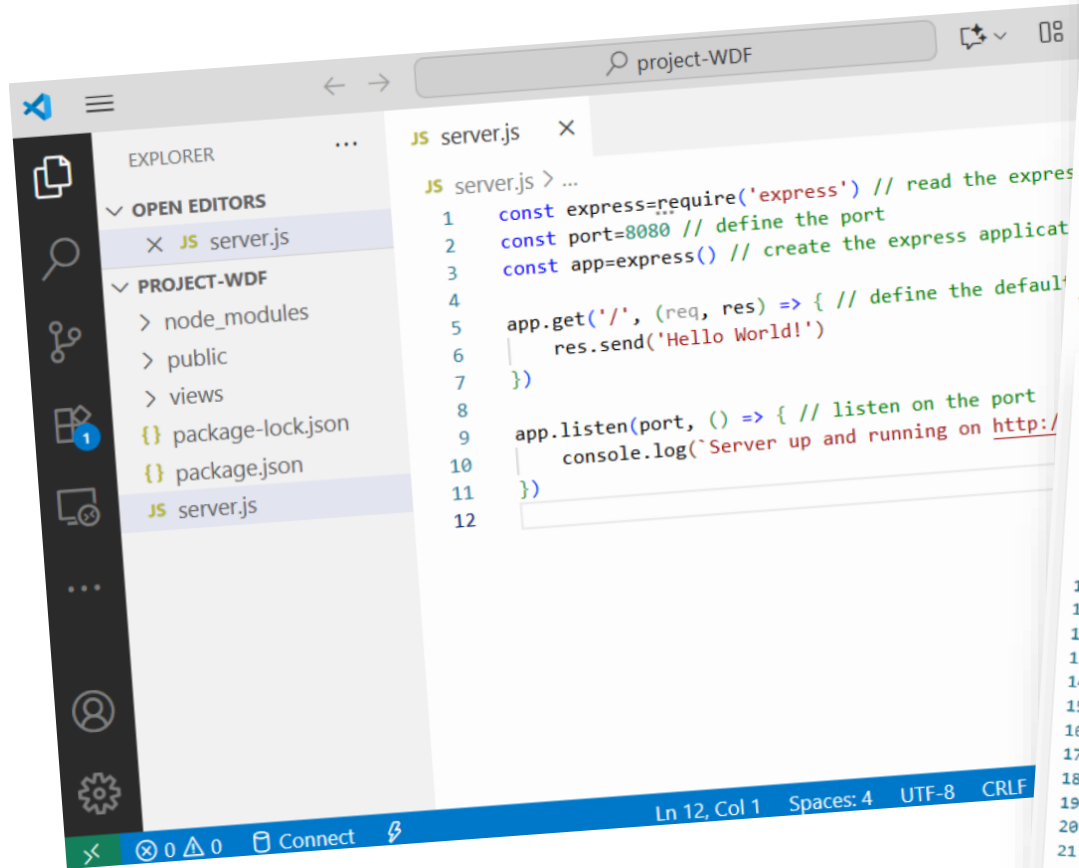
For PC, choose the correct processor and download the .msi installer:

Or get a prebuilt Node.js® for Windows running a x64 architecture.

[Windows Installer \(.msi\)](#) [Standalone Binary \(.zip\)](#)

3. My way

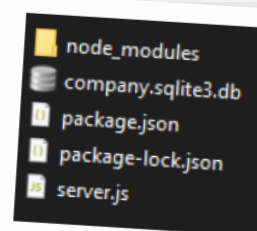
6. Learning activities: Workshops



The screenshot shows a web browser window at the top with the address bar displaying 'project-WDF'. Below the browser, the VS Code editor is open with a file named 'server.js'. The Explorer sidebar on the left shows the project structure for 'PROJECT-WDF', including 'node_modules', 'public', 'views', 'package-lock.json', 'package.json', and 'server.js'. The code in 'server.js' is as follows:

```
1  const express=require('express') // read the express
2  const port=8080 // define the port
3  const app=express() // create the express applicat
4
5  app.get('/', (req, res) => { // define the default
6      res.send('Hello World!')
7  })
8
9  app.listen(port, () => { // listen on the port
10     console.log(`Server up and running on http://
11
12
```

Modify the code to add a new route `"/listcomputers"` that will run the code to get the list of computers from the database.



```
1  const express=require('express') // load the express package into the "express" variable
2  const app=express() // create the express application in the "app" variable
3  const port=3000 // define the port that will be used by the server
4
5  // add the SQLite3 package to this server app
6  const sqlite3=require('sqlite3') // load the sqlite3 package into the "sqlite3" variable
7  const db=new sqlite3.Database('company.sqlite3.db') // load the database and put it content in the "db" variable
8
9  // define the default route "/"
10 app.get('/', (req, res) => {
11     res.send('Hello World!')
12 })
13
14 // define a new route "/listcomputers"
15 app.get('/listcomputers', (req, res) => {
16     db.all('SELECT * FROM Computer', (error, theComputers) => { // run a SQL query
17         if (error) { // if there is an error, send an error message
18             res.send('Sorry, there was an ERROR.')
19         } else { // no error, we can send back the result of the SQL query
20             res.send(theComputers) // send the RAW computers' list
21         }
22     })
23 })
24
25 // listen for incoming requests onto the port and display a message into the console (terminal)
26 app.listen(port, () => {
27     console.log(`Example app listening on port ${port}...`) // be sure to use backticks on this line!
28 })
```



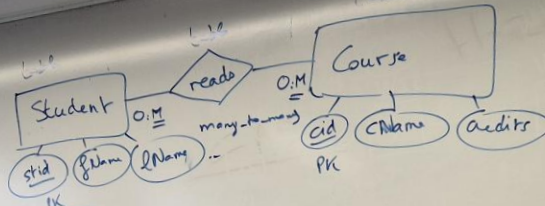
3. My way

7. Learning activities: Labs

- Computer room for labs
- Be progressive, step-by-step
- Make a pause in the middle of the session
- Answer questions, give feedback, grade the lab presentation
- Help students who have difficulties
- **Be positive!**

3. My way

7. Learning activities: Labs



Student Reads Course

R	FK Srcid	FK stid	FK cid
1	11111	7	
2	11111	23	
3	11045	23	
4	11111	34567	

many-to-many
School/Has/Department

Rate	sid	did
1	1	23
2	1	123
3	1	456
4	2	123

HELP

Liban v

Abbe/Jalle v

PR

M.
Ted

C:\> git -v

C:\> git --version

/opt/homebrew/bin/git

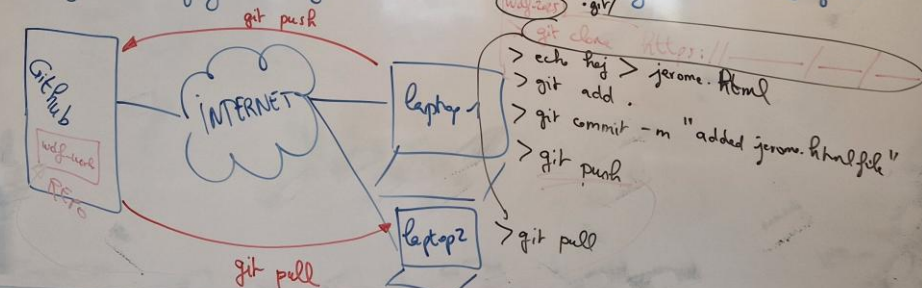
2.51.0

brew install git

23.9.15

git config --global user.name "jlandre72"

git config --global user.email "jerome.landre@ju.se"



3. My way

7. Learning activities: Labs

TGWK12 – Web Development Fundamentals



Lab 3: Node, Express, Handlebars

This lab session is dedicated to using Node, Express, and Handlebars in your project. You look at the lectures about Node, Express on Canvas. During the labs, a "projects and skills" database of your project MUST use YOUR (OWN) DATA!

1) Node, Express, Handlebars

During the project workshop, you created your project directory and populated it with SQLite3. We will start with the existing code and adapt it to use Handlebars in our project.

The starting directory structure is the one from the project workshop:

- a) You have a "public/" directory containing "css/" and "img/" subdirectories. In these, you will respectively put your css, and image files. The "public/" directory is declared as "static" in your Express application meaning that it can be publicly accessed.
- b) You also have a "views" directory where you find your HTML pages.
- c) The "server.js", "package.json" and "package-lock.json" files are directly in the project directory (not in any subdirectory).

This structure needs to change to use handlebars in your project.

- 1) Install the package "express-handlebars" in your project directory using
- 2) Add the code to use "express-handlebars" in your code:

```
const { engine } = require('express-handlebars') // load the handlebars
```

TGWK12 – Web Development Fundamentals



Lab 5: Login, bcrypt, and sessions

This lab session is dedicated to using sessions and hashed password in your project. We will continue with the "projects and skills" database in this lab. To try and test some code, you can use the data I provide for the labs, but in the end, the database of your project MUST use YOUR (OWN) DATA!

1) Hashed password

During lab 4, you created a login form and displayed the information received on the web. We will process the information, test if it is valid, and react accordingly. We need a password (as clear text for now) in our code, it must be done in the variable a

```
//--- DEFINE VARIABLES AND CONSTANTS
const port=8080
const app=express()
const adminPassword='wdf#2025'
```

We want to hash the password using the bcrypt package. You must install the bcrypt package loaded at the beginning of your "server.js" file.

```
const bcrypt=require('bcrypt')
```

We will hash this password using a function (you can put it at the end of your server.js file)

```
function hashPassword(pw, saltRounds) {
  bcrypt.hash(pw, saltRounds, function(err, hash) {
    if (err) {
      console.log('---> Error hashing password:', err);
    } else {
      console.log('---> Hashed password: ${hash}');
    }
  });
}
```

TGWK12 – Web Development Fundamentals



Lab 6: CRUD on one table and more...

This lab session is dedicated to allowing the administrator - that can be hardcoded in your application (no need to put the administrator in the table "users", especially if you do not have one) - to use CRUD on one table (I will take the "projects" table in this lab. We consider you did everything until the end of lab 5 because we will continue with the "projects and skills" database in this session. To try and test some code, you can use the data I provide for the labs, but in the end, I recall that the database of your project MUST use YOUR (OWN) DATA!

1) Making each project clickable

Follow the instructions and code given in lecture 18 to be able to make a project clickable and create a new handlebar to show details about this project.

Try your code and click on a project to get detailed information about it.

2) CRUD operations on the projects

2.1) Modifying "projects.handlebars"

To allow the CUD operations, we must ensure that the user is an admin. Follow the instructions and code

3. My way

8. The use of



6) THE USE OF AI

- You are allowed to use AI and code generation for **your project, seminars/workshops and labs** but **not for the written exam**
- **BUT** you must use AI the **smart way**:
 - Copy and paste is not sufficient!
 - You must understand the code proposed by AI
 - You must know where it goes in your files
 - You must **cite the source**, for instance

```
// code generated by ---ChatGPT---
    [... generated code here ...]
// end of code by ---ChatGPT---
```
- AI is a powerful tool when you know how to use it!
- It can help you to **LEARN**

3. My way

8. The use of



6) THE USE OF AI

- More information on AI:

<https://ju.se/portal/educate/en/guides/artificial-intelligence/>
<https://guides.library.ju.se/sourcewise/AI-tools>

”AI-generated responses may sound convincing, but it is important to remember that AI does not truly understand your subject. It generates text based on patterns in large datasets, predicting the most likely next word rather than reasoning like a human.

Even if the AI provides sources, you must always verify them. Check facts, figures, quotes, and references against credible sources to ensure accuracy. Please note that the AI may not be trained on the latest research and may therefore not provide up-to-date information.”

3. My way

9. Canvas

Dashboard

Account

Dashboard

Courses

Calendar

Inbox

History

Commons

Help

Course evaluation

TIGN13 T5031
VT2026

TWIP17 T5053
VT2026

TDRK16 T6034
VT2026

Operating Systems HT2025
TOSK18 T4233
HT2025

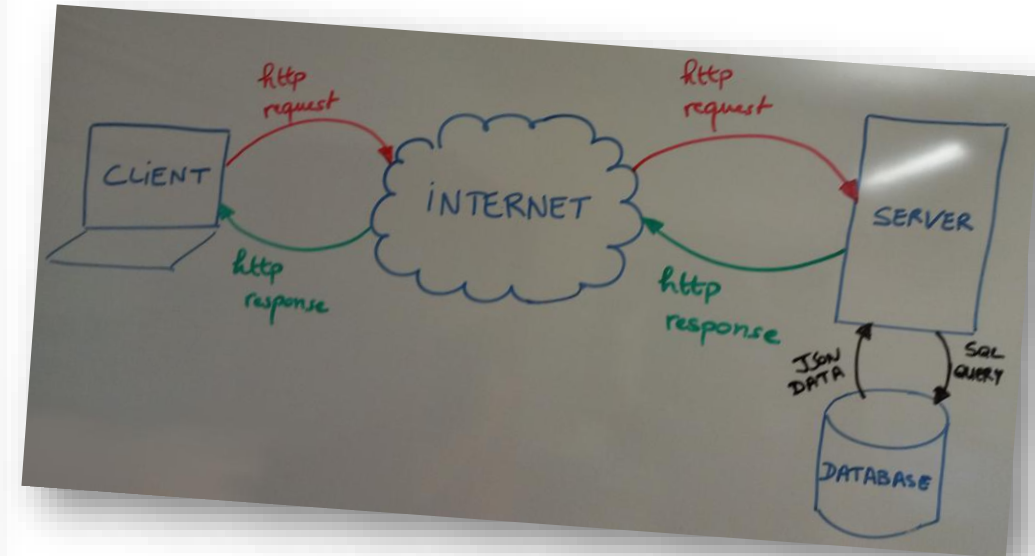
SWELANG B2/C1 part 2 for staff ...
SWLB2C1 P2 STAFF
HT2025

Future Educator - Today
FET
HT2025

Web Development Fundamentals ...
TGWK12 T4216
HT2025

Examensarbete i Datateknik VT2...
TETP10 T5026
VT2025

Intelligent Mobile Systems VT2025
TIGN13 T5031
VT2025



3. My way

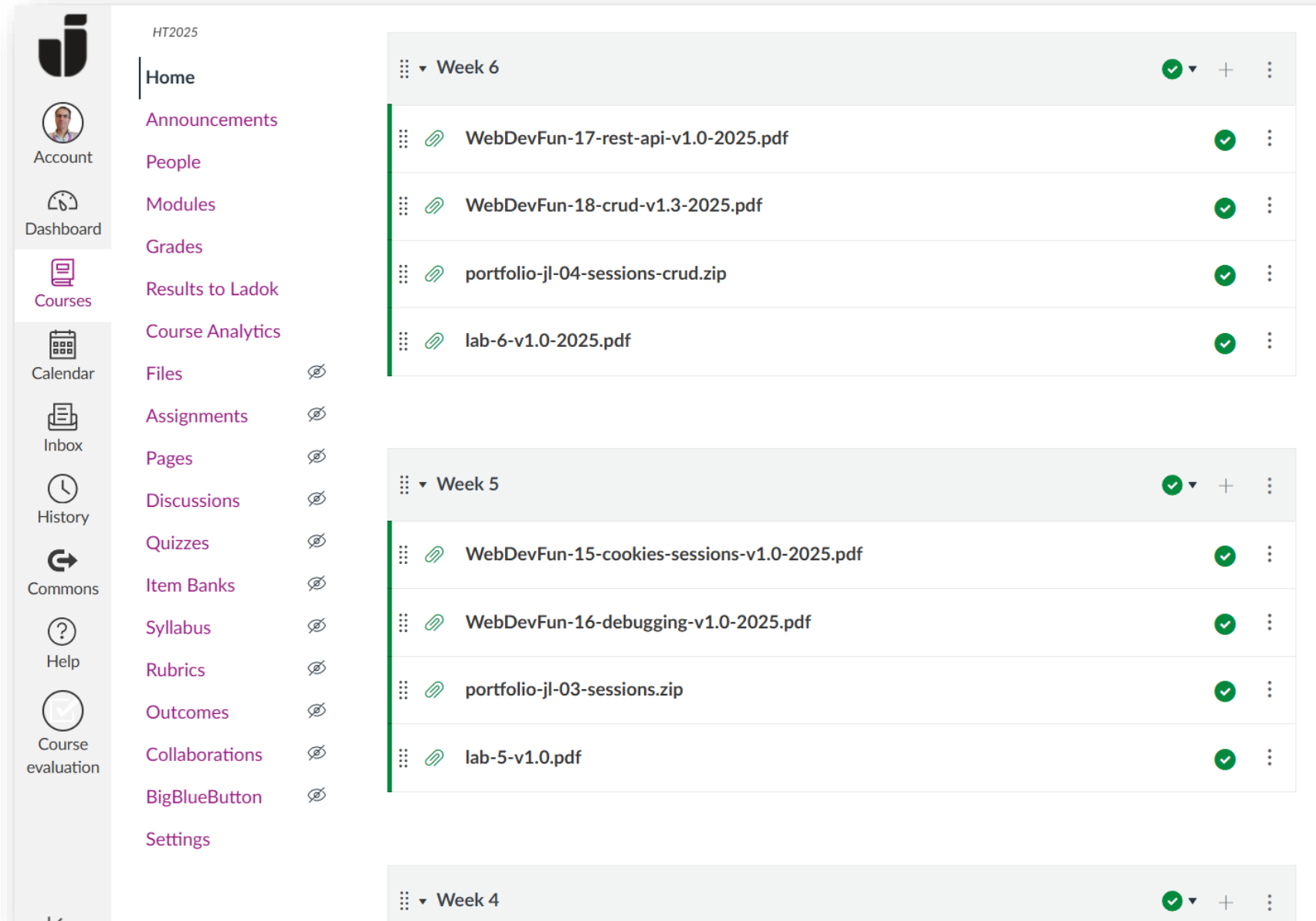
9. Canvas

The screenshot displays the Canvas LMS interface for the course HT2025. The sidebar on the left contains navigation links: Home, Announcements, People, Modules, Grades, Results to Ladok, Course Analytics, Files, Assignments, Pages, Discussions, Quizzes, Item Banks, Syllabus, Rubrics, Outcomes, Collaborations, BigBlueButton, and Settings. The main content area shows the course structure with expandable sections. The 'General information' section is expanded, showing a list of files: tgwk12-syllabus-v1.0.pdf, tgwk12-software-v1.0.pdf, schedule-labs-v1.0-2025.pdf, tgwk12-study-guide-v1.4.pdf, and tgwk12-grading-criteria-project-v1.2.pdf. The 'Week 7' section is also expanded, showing files: WebDevFun-19-pagination-upload-images-v1.3-2025.pdf, WebDevFun-20-security-tricks-v1.0-2025.pdf, and WebDevFun-21-project-submission-v1.0-2025.pdf. The 'Week 6' section is partially visible at the bottom.

Section	File Name	Status
General information	tgwk12-syllabus-v1.0.pdf	✓
	tgwk12-software-v1.0.pdf	✓
	schedule-labs-v1.0-2025.pdf	✓
	tgwk12-study-guide-v1.4.pdf	✓
	tgwk12-grading-criteria-project-v1.2.pdf	✓
Week 7	WebDevFun-19-pagination-upload-images-v1.3-2025.pdf	✓
	WebDevFun-20-security-tricks-v1.0-2025.pdf	✓
	WebDevFun-21-project-submission-v1.0-2025.pdf	✓
Week 6		

3. My way

9. Canvas

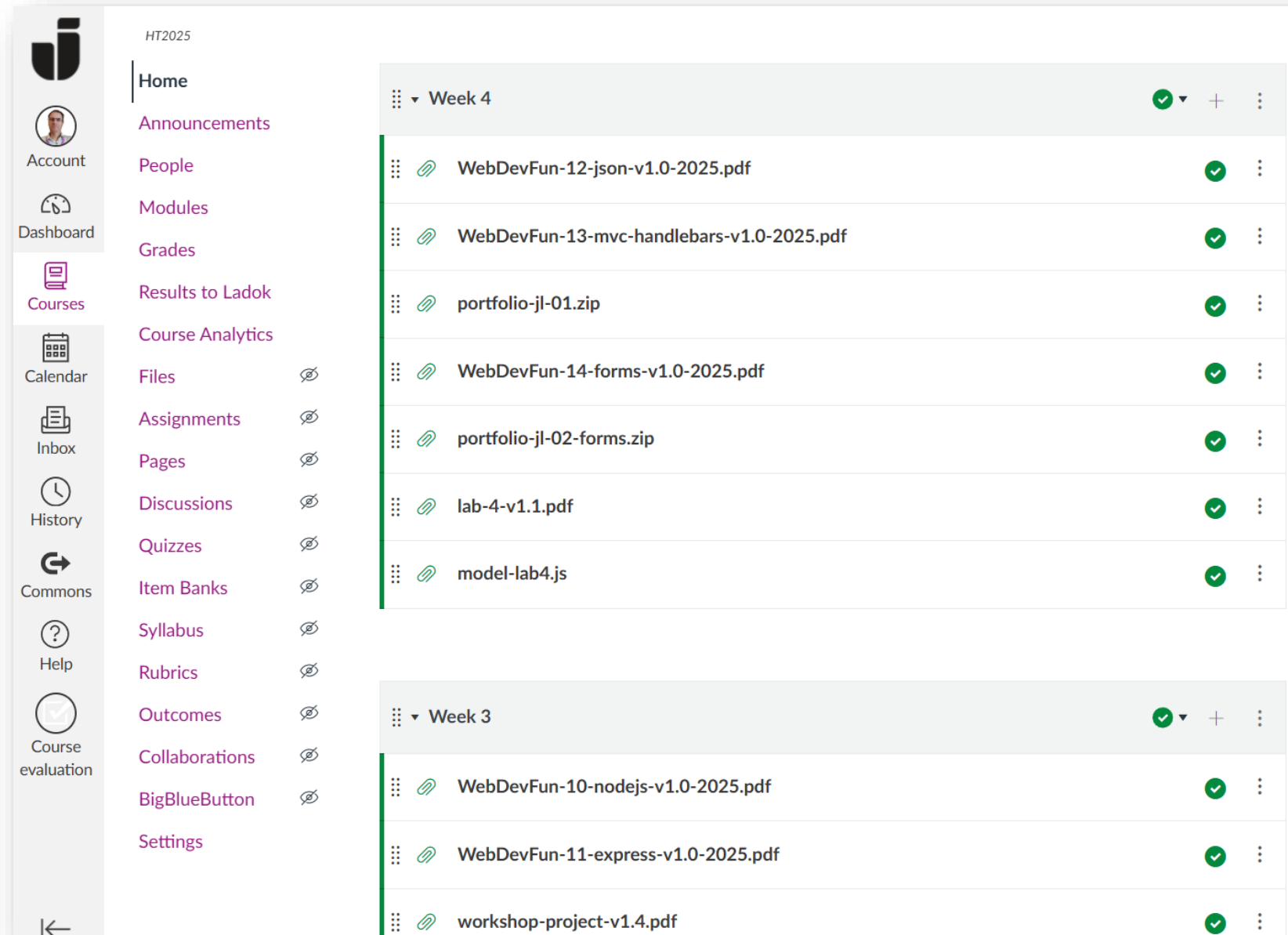


The screenshot displays the Canvas LMS interface for a course titled 'HT2025'. The left sidebar contains navigation links: Home, Announcements, People, Modules, Grades, Results to Ladok, Course Analytics, Files, Assignments, Pages, Discussions, Quizzes, Item Banks, Syllabus, Rubrics, Outcomes, Collaborations, BigBlueButton, and Settings. The main content area shows a list of files organized by week. Each file entry includes a download icon, the file name, a status indicator (green checkmark), and a more options menu (three dots).

Week	File Name	Status
Week 6	WebDevFun-17-rest-api-v1.0-2025.pdf	✓
Week 6	WebDevFun-18-crud-v1.3-2025.pdf	✓
Week 6	portfolio-jl-04-sessions-crud.zip	✓
Week 6	lab-6-v1.0-2025.pdf	✓
Week 5	WebDevFun-15-cookies-sessions-v1.0-2025.pdf	✓
Week 5	WebDevFun-16-debugging-v1.0-2025.pdf	✓
Week 5	portfolio-jl-03-sessions.zip	✓
Week 5	lab-5-v1.0.pdf	✓
Week 4		

3. My way

9. Canvas



HT2025

Home

Announcements

People

Modules

Grades

Results to Ladok

Course Analytics

Files

Assignments

Pages

Discussions

Quizzes

Item Banks

Syllabus

Rubrics

Outcomes

Collaborations

BigBlueButton

Settings

Week 4

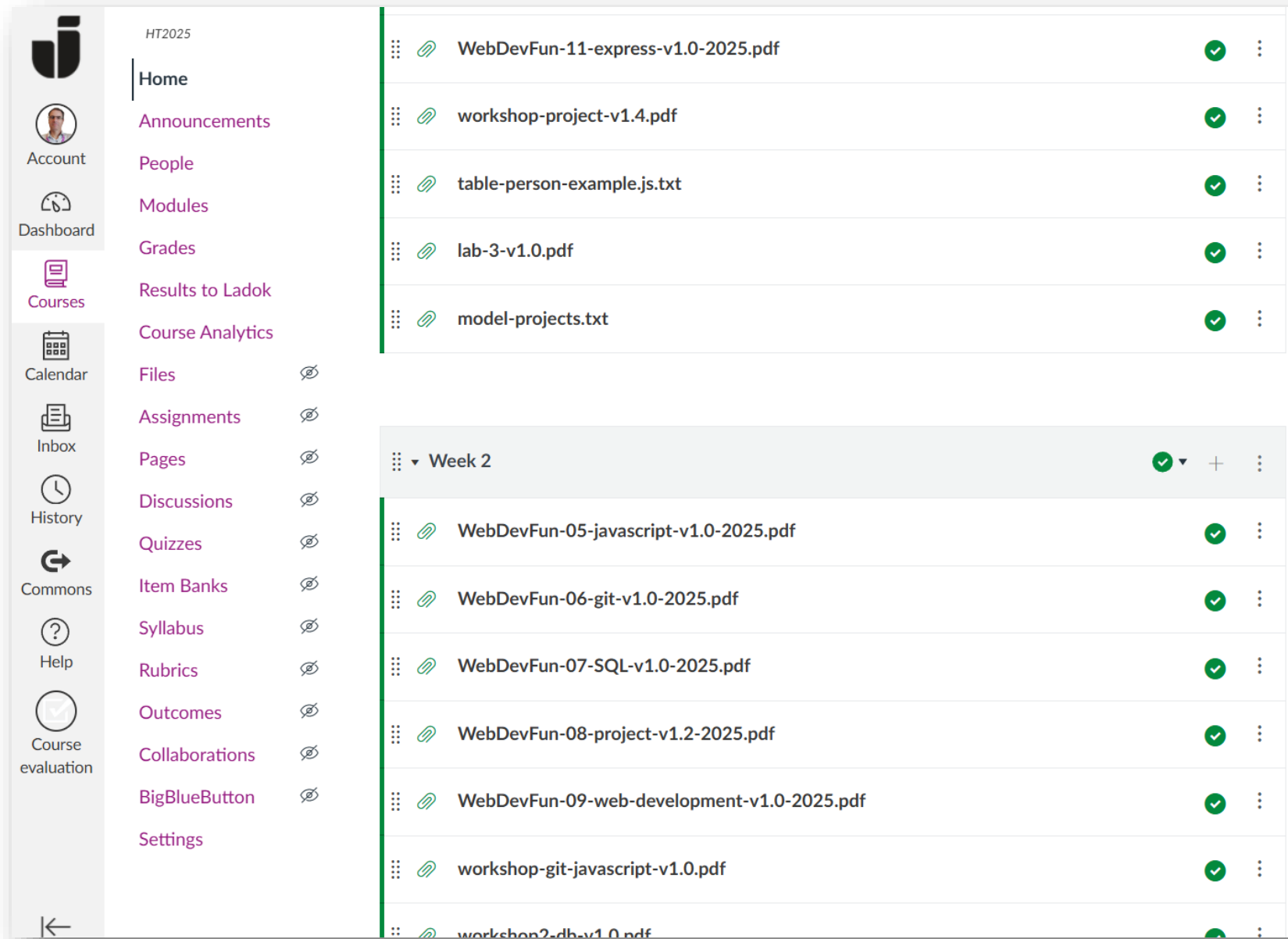
- WebDevFun-12-json-v1.0-2025.pdf
- WebDevFun-13-mvc-handlebars-v1.0-2025.pdf
- portfolio-jl-01.zip
- WebDevFun-14-forms-v1.0-2025.pdf
- portfolio-jl-02-forms.zip
- lab-4-v1.1.pdf
- model-lab4.js

Week 3

- WebDevFun-10-nodejs-v1.0-2025.pdf
- WebDevFun-11-express-v1.0-2025.pdf
- workshop-project-v1.4.pdf

3. My way

9. Canvas



HT2025

Home

Announcements

People

Modules

Grades

Results to Ladok

Course Analytics

Files

Assignments

Pages

Discussions

Quizzes

Item Banks

Syllabus

Rubrics

Outcomes

Collaborations

BigBlueButton

Settings

WebDevFun-11-express-v1.0-2025.pdf

workshop-project-v1.4.pdf

table-person-example.js.txt

lab-3-v1.0.pdf

model-projects.txt

Week 2

WebDevFun-05-javascript-v1.0-2025.pdf

WebDevFun-06-git-v1.0-2025.pdf

WebDevFun-07-SQL-v1.0-2025.pdf

WebDevFun-08-project-v1.2-2025.pdf

WebDevFun-09-web-development-v1.0-2025.pdf

workshop-git-javascript-v1.0.pdf

workshop2-dbv1.0.pdf

3. My way

9. Canvas




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Item Banks

Syllabus

Rubrics

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BigBlueButton

Settings

▼ Week 1









WebDevFun-00-introduction-v1.1-2025.pdf







WebDevFun-01-internet-http-v1.0-2025.pdf







client-server-http-request-response.jpg







WebDevFun-02-html-v1.0-2025.pdf







html-dom-tree-1.jpg







html-dom-tree-2.jpg







semantic-tags-in-html.jpg







mandatory-attributes-html.jpg







WebDevFun-03-css-v1.1-2025.pdf







WebDevFun-04-databases-1.0.pdf







table-person.jpg

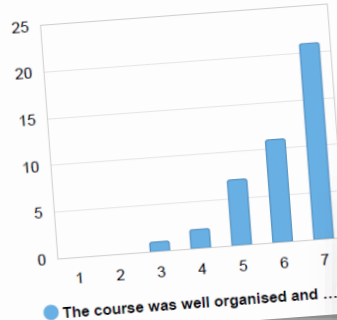




4. Conclusion

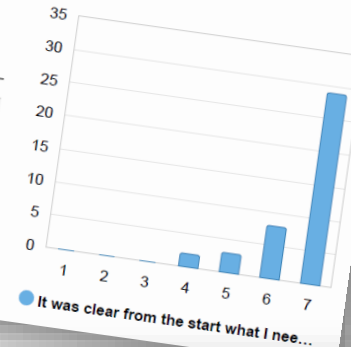
The course was well organised and structured.
1 = Do not agree at all
7 = Agree completely

	Number of responses
1	0 (0.0%)
2	0 (0.0%)
3	1 (2.4%)
4	2 (4.8%)
5	7 (16.7%)
6	21 (50.0%)
7	42 (100.0%)
Total	



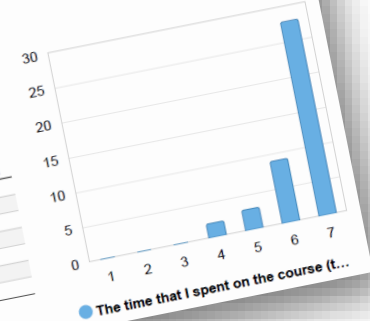
It was clear from the start what I needed to do to achieve a certain grade.
1 = Do not agree at all
7 = Agree completely

	Number of responses
1	0 (0.0%)
2	0 (0.0%)
3	0 (0.0%)
4	0 (0.0%)
5	0 (0.0%)
6	2 (4.8%)
7	3 (7.1%)
Total	8 (19.0%)
	29 (69.0%)
	42 (100.0%)



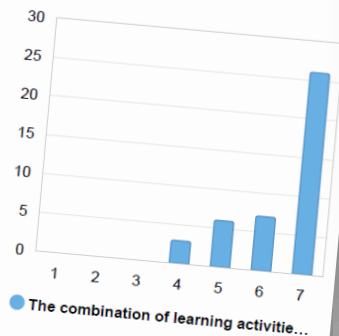
The time that I spent on the course (timetabled hours and independent study) corresponds to the number of credits that the course is worth. Please feel free to comment.
1 = Do not agree at all
7 = Agree completely

	Number of responses
1	0 (0.0%)
2	0 (0.0%)
3	0 (0.0%)
4	2 (4.8%)
5	3 (7.1%)
6	9 (21.4%)
7	28 (66.7%)
Total	42 (100.0%)



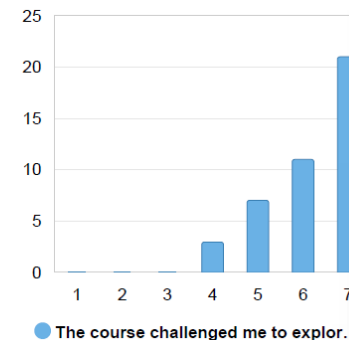
The combination of learning activities (for example lectures, exercises, seminars, projects, laboratory sessions and independent study) encouraged me to do my absolute best. Please feel free to comment.
1 = Do not agree at all
7 = Agree completely

	Number of responses
1	0 (0.0%)
2	0 (0.0%)
3	0 (0.0%)
4	0 (0.0%)
5	3 (7.1%)
6	6 (14.3%)
7	7 (16.7%)
Total	26 (61.9%)
	42 (100.0%)



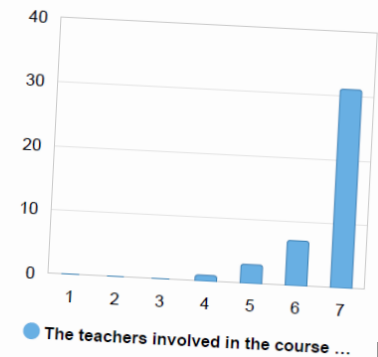
The course challenged me to explore ideas and concepts in depth.
1 = Do not agree at all
7 = Agree completely

	Number of responses
1	0 (0.0%)
2	0 (0.0%)
3	0 (0.0%)
4	3 (7.1%)
5	7 (16.7%)
6	11 (26.2%)
7	21 (50.0%)
Total	42 (100.0%)



The teachers involved in the course managed to make the subject content understandable.
1 = Do not agree at all
7 = Agree completely

	Number of responses
1	0 (0.0%)
2	0 (0.0%)
3	0 (0.0%)
4	0 (0.0%)
5	1 (2.4%)
6	3 (7.1%)
7	7 (16.7%)
Total	31 (73.8%)
	42 (100.0%)

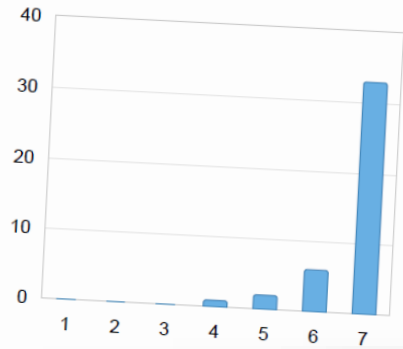


4. Conclusion

I was able to get in contact with the teachers of the course when needed.

1 = Do not agree at all
7 = Agree completely

	Number of responses
1	0 (0.0%)
2	0 (0.0%)
3	0 (0.0%)
4	1 (2.4%)
5	2 (4.8%)
6	6 (14.3%)
7	33 (78.6%)
Total	42 (100.0%)

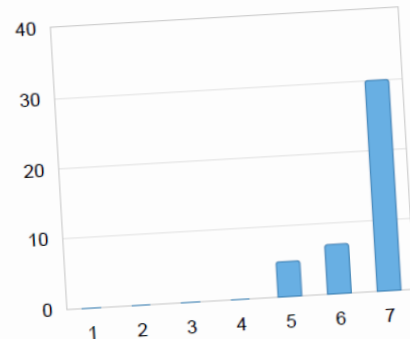


● I was able to get in...

The examination methods used were relevant in relation to the intended learning outcomes.

1 = Do not agree at all
7 = Agree completely

	Number of responses
1	0 (0.0%)
2	0 (0.0%)
3	0 (0.0%)
4	0 (0.0%)
5	5 (11.9%)
6	7 (16.7%)
7	30 (71.4%)
Total	42 (100.0%)

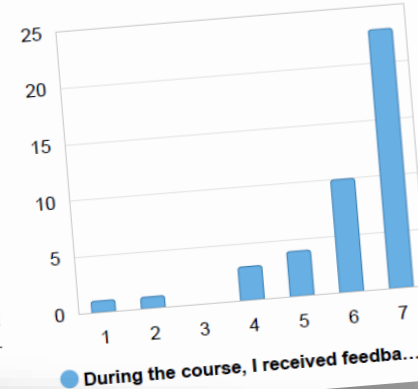


● The examination methods used were...

During the course, I received feedback (for example from teachers, students and learning activities) that contributed to my learning.

1 = Do not agree at all
7 = Agree completely

	Number of responses
1	1 (2.4%)
2	1 (2.4%)
3	3 (7.1%)
4	4 (9.5%)
5	10 (23.8%)
6	23 (54.8%)
7	42 (100.0%)
Total	42 (100.0%)

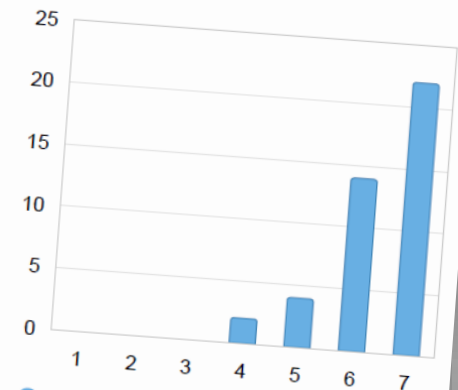


● During the course, I received feedback...

My opinion of the course as a whole.

1 = Not at all satisfied
7 = Very satisfied

	Number of responses
1	0 (0.0%)
2	0 (0.0%)
3	0 (0.0%)
4	0 (0.0%)
5	2 (4.8%)
6	4 (9.5%)
7	14 (33.3%)
Total	22 (52.4%)
	42 (100.0%)



● My opinion of the course as a whole....

4. Conclusion

What about the course do you think is most valuable to keep as it is?

Definitely keep Jerome, but give him more help as well, because it's not nice putting everything on him, and by getting people to help him, I advise they be on par with his knowledge.

The project, it was very fun to code and a very in depth assignment. Also, of course, the teacher, Jerome and his presentations!

Jerome's car plates :)

Jerome as lecturer. He made everything simple and understandable, and I felt that he wanted us to pass the course, instead of examining us on a hard level. I also feel that the exam was easy/hard enough in relation to the time to study and it is good the main focus is on our project and not the exam.

The Labs have been very helpful with gradually working on my website. The lectures were well structured with a good balance between code demonstrations and theory. Having to get help from the lab assistants each week, helped me to understand my work and solve problems quicker.

The Labs and other activities being optional so we can focus in the main project

Jerome is an amazing teacher and really engaged in the course and really helpful!! The lectures were super good and Jerome made it interesting and made students feel good about the course and get the help that was needed. Jerome 10/10. The material we had was really good and we got everything we needed.

The instructor's clear and dedicated teaching makes the course very engaging and easy to follow.

Everything

4. Conclusion

What about the course do you think is most valuable to keep as it is?

I really enjoyed your lectures. You are my favorite teacher of all time. Liked everything
Jérôme

The teacher! He was really good and funny and talked about hard things multiple times which made it easier to understand and more motivating to actually learn it.

Jérôme is an amazing teacher! He's fun and engaging and he can explain even tougher topics with ease. The workshops are also super valuable especially the one setting up your project

The project was really interesting to do and on a very good level of difficulty. The lectures were all good and added onto the understanding for the different parts of the project

The teacher, Jerome is the first teacher who have made me understand programming.

The interactive workshops at JTH towers were very useful. Groups of fewer students made me more inclined to ask questions about the material.

i liked working on the project but it felt messy sometimes with the code snippets from the lab sessions

It was easy to understand most of the course. The project was clear in what was expected and had the right level of difficulty for each grade.

The project as the big examination part is smart. And keep jerome please, he engages the whole class with questions and jokes. This makes the lectures fun and engaging so all keep focus.

The whole project we had to do is super valuable. Nothing is better than actually making a project to learn something.

Learning order with lectures and final project

4. Conclusion

What about the course do you think is most valuable to keep as it is?

The weekly lab guidance is very useful and is also an opportunity to push us to complete the task.

-
How fun the teaching is!!

The project

I really liked how Jerome encouraged and motivated the whole class to have fun. I really liked his lectures because it was pretty easy to follow and not too hard to understand and it was pretty funny as well. I really liked that we did get the code, because in the previous coding courses only the rare basics got explained and we had to figure out the rest ourselves. But this is almost impossible, if you have no prior coding knowledge. That is why I love this course, because we got a very solid base to work with in the future ourselves

The labs and seminars

4. Conclusion

Score points with the students

- **One smile = one point**
- Surprise the students
- Use humor (sparsely)
- Make them curious
- Make them have success
- Be holistic: **savoir-faire** et **savoir-être**
- Make a difference
- **Be positive!**

Communication

- Be consistent
- Create your branding
- Answer students' questions
- Answer students' emails
- Send an announcement on Canvas every week to keep them on track
- **Be positive!**

Make the students **RESPONSIBLE** for **their** learning!

4. Conclusion

- Be positive!  • Take **EDUCATE** courses! <https://ju.se/EDUCATE>
- Everyone can win the  prize!
- JTH offers a nice environment to teach: 
- You need a little bit of 
- And more, much more than this,  **YOUR** 



Thank you, Frank!

Source: <https://www.youtube.com/watch?v=w019MzRosmk>

Jérôme Landré, Assistant Professor
plan B expert <https://jlandre72.github.io/>



Jönköping University 

Université de Reims Champagne-Ardenne 

Université de Bourgogne
Ph.D. in Image Processing 

**Be positive,
inclusive and
flexible!**

- Research: Image Processing
- Teaching: Databases, Operating Systems, Web Development, Image Processing

Car plate
collector



Tourism (Sweden)



Boardgames



Thank you for the AWARD!

Any questions?



JÖNKÖPING UNIVERSITY